Upasana Sridhar

Ph.D. Candidate, Electrical and Computer Engineering, Carnegie Mellon University

upasana.sridhar@gmail.com
GitHub: UpasanaSridhar
Phone: 508–667–4322

Research Summary Making the reasoning behind high-performance code explicit, reusable, and checkable through performance patterns, abstractions that preserve them, and contracts that verify implementations use the hardware as intended. Broader goal: programming models in which humans and AI systems can state, share, and check performance expectations.

Research Interests: High-performance computing, performance modeling, programming abstractions, formal verification.

Education

Carnegie Mellon University Aug. 2019 – Present
Ph.D. Electrical and Computer Engineering, expected December 2026
M.S. Electrical and Computer Engineering, December 2026
QPA: 3.66

PES University, Bengaluru, India Aug. 2015 – May 2019
B.Tech. Computer Science, 2019
GPA: 9.08/10.00

Honors and Awards

- **International HPC Summer School Selection** 2026
  Selected as one of approximately 100 early-career researchers worldwide for an intensive international program in high-performance and scientific computing, with opportunities to engage with a global community of computational scientists.
- **IPDPS Ph.D. Forum Selection** 2025
  One of 26 students selected to present dissertation research at the IEEE International Parallel and Distributed Processing Symposium Ph.D. Forum.
- **Proposal accepted to AMD Pervasive AI Contest** 2024
  Received an AMD Ryzen Mini-PC to study the performance tradeoffs of the AIE accelerator.
- **CMU ECE Commitment to DEI Award** 2024
  Recognized contributions to inclusion, belonging, and community-building within the Electrical and Computer Engineering department.
- **CMU CIT Dean's Ph.D. Fellowship Award** 2019
  Fellowship covering tuition and stipend for the first year of Ph.D. study.

Publications

Peer-Reviewed Journal Articles

- **Upasana Sridhar**, Nicholai Tukanov, Elliott Binder, Tze Meng Low, Scott McMillan, and Martin D. Schatz. "SMaLL: Software for Rapidly Instantiating Machine Learning Libraries." *ACM Transactions on Embedded Computing Systems (TECS)*, 2024.

- Sergio Barrachina, Adrián Castelló, Manuel F. Dolz, Tze Meng Low, Héctor Martínez, Enrique S. Quintana-Ortí, **Upasana Sridhar**, and Andrés E. Tomás. "Reformulating the Direct Convolution for High-Performance Deep Learning Inference on ARM Processors." *Journal of Systems Architecture (JSA)*, 2023.

Peer-Reviewed Conference and Workshop Papers

- **Upasana Sridhar**, Elliott Binder, and Tze Meng Low. "Gen-AI in a Bottle: Experiments with LLMs to Generate HPC Kernels." *IEEE IPDPS Workshops (IPDPSW)*, 2025.
- Yuttapichai Kerdcharoen, **Upasana Sridhar**, Orathai Sangpetch, and Tze Meng Low. "Linear Algebra Approach for Directed Triad Counting and Enumeration." *SC24 Workshops*, pp. 718–726, 2024.
- Yuttapichai Kerdcharoen, **Upasana Sridhar**, and Tze Meng Low. "Exploiting Fusion Opportunities in Linear Algebraic Graph Query Engines." *IEEE High Performance Extreme Computing Conference (HPEC)*, 2023.
- Tze Meng Low, Yuejie Chi, James C. Hoe, Swarun Kumar, Akarsh Prabhakara, Laixi Shi, **Upasana Sridhar**, Nicolai Tukanov, Chengyue Wang, and Yuchen Wu. "Zoom Out: Abstractions for Efficient Radar Algorithms on COTS Architectures." *IEEE International Symposium on Phased Array Systems & Technology (PAST)*, 2022.
- **Upasana Sridhar**, Tze Meng Low, and Martin D. Schatz. "Fusing Non Element-wise Layers in DNNs." *IEEE High Performance Extreme Computing Conference (HPEC)*, 2021.
- Ariful Azad, Mohsen Mahmoudi Aznaveh, Scott Beamer, Maia P. Blanco, Jinhao Chen, Luke D'Alessandro, Roshan Dathathri, Tim Davis, Kevin Deweese, Jesun Firoz, **Upasana Sridhar**, Tze Meng Low, and Martin D. Schatz. "Evaluation of Graph Analytics Frameworks Using the GAP Benchmark Suite." *IEEE International Symposium on Workload Characterization (IISWC)*, 2020.
- Daniele G. Spampinato, **Upasana Sridhar**, and Tze Meng Low. "Linear Algebraic Depth-First Search." *ACM SIGPLAN ARRAY Workshop*, pp. 93–104, 2019.
- **Upasana Sridhar**, Maia P. Blanco, Rahul Mayuranath, Daniele G. Spampinato, Tze Meng Low, and Scott McMillan. "Delta-Stepping SSSP: From Vertices and Edges to GraphBLAS Implementations." *IEEE IPDPS Workshops*, 2019.
- Tze Meng Low, Daniele G. Spampinato, Anurag Kutuluru, **Upasana Sridhar**, Doru Thom Popovici, Franz Franchetti, and Scott McMillan. "Linear Algebraic Formulation of Edge-Centric k-Truss Algorithms with Adjacency Matrices." *IEEE High Performance Extreme Computing Conference (HPEC)*, 2018.

Technical Reports, Briefs, and Other Publications

- Scott McMillan, Jay Palat, Oren Wright, **Upasana Sridhar**, Tze Meng Low, Nicolai Tukanov, Pankti Shah, and Navya Chandra. "PHITE: Portable High-Performance Inference at the Tactical Edge." 2022.
- Scott McMillan, Tze Meng Low, Oren Wright, **Upasana Sridhar**, and Nicolai Tukanov. "Member Brief: SEI, AI Division, Advanced Computing Lab." 2022.
- Scott McMillan, Elliott Binder, Mark Blanco, Paul Brower, Franz Franchetti, James C. Hoe, Tze M. Low, Peter Oostema, Fazle Sadi, Daniele Spampinato, and **Upasana Sridhar**. "Spiral/AIML: Frontiers of Graph Processing in Linear Algebra." 2022.

Research Experience

Carnegie Mellon University — Graduate Research Assistant Aug. 2019 – Present
Advisor: Prof. Tze Meng Low
Thesis Title: Designing Extensible and Portable Performance Libraries for CNNs
Reduced the specialization burden of implementing a CNN library by reusing performance design across operators and hardware; supports 130 ONNX operators with 36 specialized microkernels organized into two dataflow patterns.

Microsoft Research — Research Intern May 2023 – Aug. 2023
Redmond, WA
Manager: Srikant Bharadwaj
Developed an analytical model for mixed-precision foundation-model inference to evaluate accuracy, latency, and throughput tradeoffs. Demonstrated that up to 20% of model weights could remain in high precision without degrading inference performance.

Microsoft Research — Research Intern May 2021 – Aug. 2021
Managers: Patrice Godefroid and Marina Polishchuk
Worked on RESTler, a stateful REST API fuzzer, using production API traffic to improve request generation. Applied anomaly detection to identify fields causing request failures and developed a model to retry failed requests using predicted field values.

Carnegie Mellon University — Research Staff Jan. 2019 – May 2019
Advisor: Prof. Tze Meng Low
Developed a code-generation framework for outer-product-like operations across applications including clustering and computational genetics. Abstracted their shared pairwise-operation-and-reduction structure to automatically generate high-performance x86 assembly microkernels.

Carnegie Mellon University — Research Intern Jun. 2018 – Aug. 2018
Advisor: Prof. Franz Franchetti
Studied formal methods for deriving provably correct graph algorithms and applied these techniques to instantiate high-performance implementations of triangle counting on GPUs and k-truss on CPUs.

Invited Talks

- **SMaLL: Portable and Rapidly Instantiable Machine Learning Libraries**
  - PES University – Nov. 2023
  - Indian Institute of Science (IISc) – Nov. 2023
  - BLIS Retreat – 2023
  - Schlumberger (SLB) – Aug. 2025
- **Composable Abstractions for Layer Fusion: Convolution, Pooling, and Beyond**
  - BLIS Retreat – 2024
- **Stories from the Field: Designing Production AI Systems**
  - PES University – Mar. 2023
- **AI for Bottle Screening**
  - CMKL University, Thailand – Jul. 2021

Conference Presentations and Posters

- **Gen-AI in a Bottle: Experiments with LLMs to Generate HPC Kernels**
  Presentation, International Workshop on Automatic Performance Tuning (IWAPT), 2025.
- **Beyond Elementwise Fusion in CNNs**
  Poster presentation, IEEE High Performance Extreme Computing Conference (HPEC), 2021.
- **Designing Reusable Performance Libraries for Edge ML**
  Poster presentation, International HPC Summer School (IHPCSS), 2026.
- **A Unified Abstraction for Efficient and Portable CNN Libraries on Edge-Devices**
  Poster presentation, IPDPS Ph.D. Forum, 2025.

Teaching and Mentoring

Courses Taught / Assisted

- **How to Write Fast Code II**
  - **Guest Instructor** – Spring 2024, Spring 2026
    Lectures on GPU Architecture and Programming; Task-based Parallelism in OpenMP.
  - **Project Mentor / Subject-Matter Expert** – Spring 2023, Spring 2024
    Provided questions and feedback during final presentations of student-led projects.
- **Computational Problem Solving for Engineers**
  - **Teaching Assistant** — Spring 2025
    Designed recitations and assignment questions, ran office hours and graded assignements.
- **How to Write Fast Code I**
  - **Guest Instructor** — Fall 2022, Fall 2023
    Delivered lectures introducing SIMD instructions, designing kernels for Convolutions.
  - **Teaching Assistant** — Fall 2020, Fall 2021
    Supervised 9 teams of 3-4 students working on the open-ended class projects. Ran office hours, graded homeworks, managed a hybrid classroom.
  - **Project Mentor / Subject-Matter Expert** — Fall 2022, Fall 2023
    Provided feedback to project teams on their designs and on their final project presentations.
- **Advanced Algorithms**
  - **Teaching Assistant** — Fall 2018
  - Project-based course
- **Nature of Code**
  - **Instructor** — Jan.–Mar. 2018
  - Student-taught course, PES University

Teaching Training

- Completed *Teaching As Research (TAR) Institute*, CMU Eberly Center, 2022. A 3-day program focused on developing skills to use data to inform teaching. Covered data sources, study design, and incorporating feedback and presenting student outcomes.
- Completed *Evidence-Based Teaching in STEM*, CMU, Spring 2024. A semester-long course designed to prepare graduate students in STEM disciplines to: (1) teach effectively and efficiently as future faculty members; (2) critically read and apply peer-reviewed, STEM-based educational research; and (3) adapt approaches from the Scholarship of Teaching and Learning (SoTL) to formatively assess student learning and iteratively improve teaching and course design. *Course Developed: How to Think in Code - Introduction to CS for non-majors*
- Completed *Future Faculty Program*, CMU Eberly Center, Spring 2026.
  A program to help graduate students and postdocs develop teaching skills and document their teaching-related professional development activities in preparation for a faculty career.

Research Mentorship

Undergraduate Students
Delaynie McMillan - Design-Space Exploration of Specialized Assembly implementations of Convolution Layers

Masters Students
Alex Bauerle and Dharun Muthaiah Natraj - Unified Library design for ML primitives on GPUs
Daichen Liu - Reducing Engineering Time to High-performance ONNX backend libraries
Sai Subha - Accelerating CNNs using fp16 instructions on ARM processors
Shreyashri Biswas - Analytical Modeling of CNN models to design high-performance assembly kernels
Navya Chandra - Specializing CNN libraries for ARM (micro)processors

Ph.D. Students
Yuttapichai Kerdcharoen - Families of Provably Correct Algorithms for Motifs in Graph Databases
Yufei Shi - Do AI-Agents dream of Peak? How to communicate performance expertise to democratize access to efficient implementations
Chenxi Wan - Formally Reasoning about Fusion using operation-level Loop Invariants.

Service

Professional Service

- **Workshop Co-Organizer** 2nd LLM4HPC Workshop, ISC High Performance, 2026.
- Program Committee Member, 1st International Workshop on Agentic AI for HPC, SC 2026
- Program Committee Member, 13th Workshop on Accelerator Programming and Directives, SC 2026.
- Program Committee Member, 1st Workshop on LLMs for HPC, SC Asia 2026.
- Program Committee Member, AI Track, International Conference on Parallel Processing (ICPP) 2025.

Departmental and University Service

- **Co-Chair**, Student Roundtable for Faculty Recruitment Interviews, CMU ECE, Spring 2023 and 2024.

- **ECE Open House Student Representative**, Carnegie Mellon University, 2020-Present.
- **DEI Student Advisory Committee**, CMU College of Engineering, November 2021 - May 2025.
- **Evaluator** Undergraduate Honors Thesis Presentations, CMU College of Engineering, April 2024, 2026.

Outreach and Broadening Participation

- CMU Tartan Bike Project: Volunteer Mechanic, Spring 2026 - Present. Provide hands-on, just-in-time instruction in bicycle maintenance; develop instructional resources addressing common repair difficulties; help organize community-responsive events.
- **Pittsburgh Women in Mathematics and Computing Conference**: Planning & Publicity Chair, one-day workshop for undergraduates in the Pittsburgh area, February 2023.
- **Get Performance With Me: Pipelining in the Real World**, Speaker, 2024 GAINS Conference for high-school gender minorities in STEM.
- **STEM Outreach, CMU College of Engineering**, Spring 2021–Present. In-class BBC Microbit programming workshops and after-school STEM experiences.
- **Skype-a-Scientist**, Spring 2023. Introductory computer graphics session for middle school students.
- **Read-a-Story (India)**: Volunteer at a literacy and reading comprehension program.

Skills

Languages: C, C++, Python, x86 Assembly, ARM Assembly, SystemVerilog, Matlab, Lean4
Platforms & Frameworks: CUDA, Triton, AMD AIE/NPU, OpenMP, MPI, Slurm, PyTorch, TensorFlow, ONNX, TVM, GraphBLAS
Tools: NVIDIA Nsight/nvprof, Linux, perf, vtune, gem5, Git, CMake
Research Expertise: Reusable ML Library Design, Analytical Performance Modeling, Heterogeneous Computing, Software-Hardware Codesign, Hardware-Aware Applied ML, Distributed Computing, GPU Programming, Formal Methods for High-Performance Code